SHARI DEANNA ANDERSON OLIVER

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PROFESSIONAL SUMMARY

An emerging Medical Physicist with a strong foundation in radiation physics, cancer research, and computational methods. Experienced in dosimetry research, imaging analysis, and health equity work through the Jamaica Cancer Research and Equity Review. Recognized for academic excellence, scientific leadership, and a commitment to improving oncology outcomes in underserved communities. Passionate about advancing radiotherapy accuracy and strengthening cancer-care access in resource-limited settings.

Research Interests: Radiation Oncology Physics; Dosimetry; Medical Imaging, Computational Modelling; Biophotonics.

EDUCATION

The University of the West Indies, Mona *2022-Present*
B.Sc. Medical Physics & Computer Science (Expected 2026)
Current GPA: 3.62/4.3* | Expected Classification: First Class Honors (pending final year results)
Relevant coursework: Advanced Practices in Medical Physics (PHYS3300), Biomedical Optics & Biomechanics (PHYS33341), Medical Radiation & Imaging (PHYS3389), Practices in Medical Physics (PHYS2200), Software Engineering (COMP2140), Computer Applications in Physics (PHYS2396)

Convent of Mercy Academy 'Alpha' *2016-2022*
High School Diploma, Honors
CSEC: 6 Grade I, 2 Grade II | CAPE: 3 Grade I, 1 Grade II

RESEARCH & PROJECTS

National Cancer Research, Diagnostic, Screening & Treatment Access *Sept 2025-Present*
Jamaica Cancer Research and Equity Review (JCRER)

- Contributing to national research assessing access, delays, and disparities in Jamaica's cancer-care pathway.
- Analyzing diagnostic timelines, treatment initiation, and systemic barriers affecting patient outcomes.

Exploration of Biophotonics for Early Cancer Detection Using Optical Spectroscopy & Hyperspectral Imaging
Oct - Nov 2025

- Investigated tissue-light interactions and spectral biomarkers for precancerous detection.
- Evaluated potential for non-invasive imaging innovations in early cancer screening.

Validation of a Step Counter Device Using Controlled Gait Experiments *Oct - Dec 2025*

- Designed controlled gait experiments to assess smartphone step-count accuracy.
- Compared devices' output against reference measurements across varying gait speeds.

Optics in Medical Imaging: Image Formation & Diagnostic Applications *Sept 2025*

- Explored fluorescence, scattering, absorption, and wavefront changes in optical imaging.
- Analyzed how system parameters influence resolution and contrast in OCT and fluorescence imaging.

A Comparison of Dosimetry Techniques in Resource-Limited Settings *Jan - May 2025*

- Assessed the feasibility and accuracy of alternative dosimetry methods.

- Evaluated reproducibility and clinical suitability for low-resource radiotherapy settings.

"Hey Gluco": AI-Integrated Smart Glucometer Prototype *Jan - May 2025*
Undergraduate Medical Physics Research
- Developed accessible glucometer prototype with preloaded strips, audio/haptic cues, and Bluetooth connectivity.
- Applied human-factors design principles to support visually impaired users.

UWIDormFinder: Campus Accommodation Search Tool *Sept 2024 - May 2025*
- Developed Java-based tool enabling dormitory search, filtering, and comparison.
- Enhanced skills in OOP, data structures, and UI logic.

Car Inventory Management System *Jan - May 2024*
- Created an inventory-management application with sales tracking and reporting features.
- Strengthened understanding of algorithms, databases, and interface design.

RELEVANT PROFESSIONAL EXPERIENCE

Summer School Coordinator & Teacher *May 2021-Aug 2025*
Trelawny Parish Library
- Taught Computer Literacy, Integrated Science, Reading, and Creative Writing to children ages 6-14, strengthening STEM engagement and foundational academic skills through interactive instruction.
- Coordinated key aspects of the annual summer programme, including scheduling, activity planning, and volunteer support, contributing to smoother programme execution and improved student participation.

STEM Tutor *April 2023-Present*
Prime Pupils Ja, Studucate International & The Hope Tutoring Project
- Taught CSEC and CAPE Physics to more than 200 students through structured online and in-person lessons, strengthening conceptual understanding, exam readiness, and confidence in STEM subjects.
- Founded Prime Pupil JA, a student-centered tutoring platform offering individualized physics instruction and revision support; developed high-yield teaching materials and tailored learning strategies to improve student outcomes.

Founder & Executive Director *Aug 2025-Present*
STEMpower Foundation
- Established a youth-led non-profit promoting STEM education and empowerment across Jamaica and the Caribbean.
- Coordinating national outreach programs, school partnerships, and mentorship initiatives for over 200 students.

Computing Adjunct Laboratory Assistant *Sept -Dec 2025*
UWI Computing Department
- Facilitating and grading labs and assisting with debugging and coursework for over 500 students across computing undergraduate courses.

Researcher *Sept 2025-Present*
Jamaica Cancer Research & Equity Review (JCRER)
- Conducting research on healthcare access disparities and presenting findings on national cancer-care challenges.

LEADERSHIP EXPERIENCE

Demonstrated progressive leadership across academic, administrative, and student-governance roles, contributing to student advocacy, STEM outreach, and institutional development at the University of the West Indies. Supported the

execution of major faculty-wide initiatives (in my role as Faculty Representative), including the **UWI Science Festival**, which brought over 150 schools from across Jamaica to the Faculty of Science and Technology for hands-on STEM exposure. Curated and assisted in delivering academic workshops, strengthened communication between over 3000 students and faculty, improved club operations, expanded community engagement, and helped shape policies and programming that enhanced the student experience and visibility of science on campus.

Roles held:

- Vice President & P.R.O, UWI Astronomy Club (2025-2026)
- Secretary & Events Coordinator, Irvine Hall Sign Language Club (2025-2026)
- Treasurer, Medical Physics Club (2025-2026)
- Committee Advisor, FST Guild Committee (2025-2026)
- Faculty of Science & Technology Representative (2024-2025)
- Faculty Cohort Secretary, UWI Mona Guild (2024- 2025)
- Peer Career Advisor (2024-2025 & 2025-2026)
- Disciplinary Committee Member, UWI Examinations Section (2024-2025)
- Mentor, FST Cosmic Companions Mentorship Programme (2024-2025 & 2025-2026)
- UWI UCJ Accreditation Member- Faculty Group (2024- 2025)
- National Youth Parliamentarian (2023-2024)
- Vice President, Medical Physics Club (2023-2024)
- Physics Department Representative, FST Guild Committee (2023-2024)
- Physics Deputy Representative, Physics Department Subcommittee (2022-2023)
- Deputy Chair, FSTGC Constitutional Advisory Committee (2022-2023)
- Secretary, Faculty of Science and Technology's Public Relations Committee (2022- 2023)
- Public Relations Officer, FST VP Properties and Special Initiatives Committee (2022- 2023)
- Course Representative: PHYS0411, PHYS0412, PHYS0421, PHYS0422, ELET1500, PHYS2296, PHYS2386, PHYS3341, PHYS3395

RELEVANT ACHIEVEMENTS & AWARDS

3× Dean's Academic Commendation Awardee | 3× Physics Honor Society Awardee | 2× Charlton Foundation Award for Academic Excellence | UWI Toronto Benefits Award Scholarship | Fancy Frenz Scholarship Award for Academic Excellence | Dean's Volunteer of the Year | Michael Neville Foundation Award | 3× Physics Honours Society Award | Regional Merit List- 6th CAPE Geography | UWI Outstanding Leadership Award | VM Foundation Master Plan Scholarship | Pinnacle Award for Excellence in Leadership | Excellence in Committee Leadership- UWI Mona Guild | UWI OSSD Outstanding Performance in Student Leadership | Betting Gaming & Lotteries Commissions Grant | 2× Ministry of Finance Tertiary Grant

RELEVANT SKILLS

- *Programming Languages:* Python, MATLAB, SQL, Java, HTML, CSS
- *Tools & Platforms:* Image J/FIJI, Excel, Git
- *Research & Analytical Skills:* Technical writing, data visualization, computational troubleshooting
- *Languages*: English (native), Spanish (proficient), Jamaican Sign Language (basic), French (basic).